AYMANE AIT DADS

Algorithm Developer Intern | ML Pipelines · Signal Processing · PyTorch

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Hiring Team - Algorithm Developer Intern
Applied Materials · Santa Clara, CA, United States

Dear Applied Materials Hiring Team,

Your posting calls for applying **data science techniques including ML and AI** to **sensor signals collected from semiconductor equipment tools** - exactly the problem class I tackled in my anomalous sound detection project, where I built a **transformer autoencoder** for unsupervised fault detection on industrial equipment signals, reaching **AUC 0.80** with zero labeled anomalies, evaluated on a DCASE-style benchmark that mirrors the multi-tool signal diversity you describe across Etch, MDP, and DDP platforms.

To connect algorithm development directly to **workflow acceleration**, I engineered an end-to-end **ML pipeline on 100K+ telemetry records** at Orange Maroc using **Random Forest and KMeans**, cutting manual analysis time by **~60%** and mapping 5 user performance profiles to commercial tiers for executive decisions. Separately, I ran structured **ablation studies** across transformer unfreezing depth, learning rate schedules, and augmentation strategies on a **428M-parameter CLIP model** fine-tuned on 250K samples - ranking **#1 on the EURECOM leaderboard** - demonstrating the systematic experimentation discipline your algorithm developers rely on.

Applied Materials sits at the exact intersection where advanced process control and data-driven algorithms determine whether the next generation of chips ships on time - the sensor signals your tools generate are the raw material for that control loop. I would welcome the chance to discuss how my background fits what your team is building in Santa Clara.

Sincerely,

AYMANE AIT DADS